

Dataset 5: Exploratory Factor Analysis

Psychological Constructs Study — 15 Variables, 4 Latent Factors

1. Research Objective

This study applies Exploratory Factor Analysis to 15 psychological variables measured across 400 individuals to uncover the latent factor structure underlying observed psychological responses. The goal is to reduce dimensionality from 15 observed variables to a smaller set of interpretable latent constructs representing distinct psychological domains, which can inform scale development and clinical assessment.

2. Variables (All Observed, No Pre-Defined DV/IV)

Variable	Hypothesized Factor
Worry_Level, Panic_Symptoms, Sleep_Disturbance	Anxiety & Distress
Sadness_Score, Hopelessness, Fatigue_Level	Depression & Hopelessness
Public_Speaking_Confidence, Social_Interaction, Assertiveness	Social Confidence
Self_Esteem, Life_Satisfaction, Coping_Ability	Resilience & Coping
Impulse_Control, Stress_Tolerance, Emotional_Awareness	Cross-loading / Mixed

3. Methodology — Exploratory Factor Analysis (EFA)

EFA is selected because the research objective is to discover latent structure rather than confirm a pre-specified model (which would call for CFA). Maximum Likelihood extraction with Varimax orthogonal rotation is used to maximize the interpretability of loadings. Factors are retained using the Kaiser criterion (eigenvalue > 1) and scree plot analysis. Adequacy of the correlation matrix for factor analysis is verified via Bartlett's test and the Kaiser-Meyer-Olkin (KMO) measure.

4. Key Results

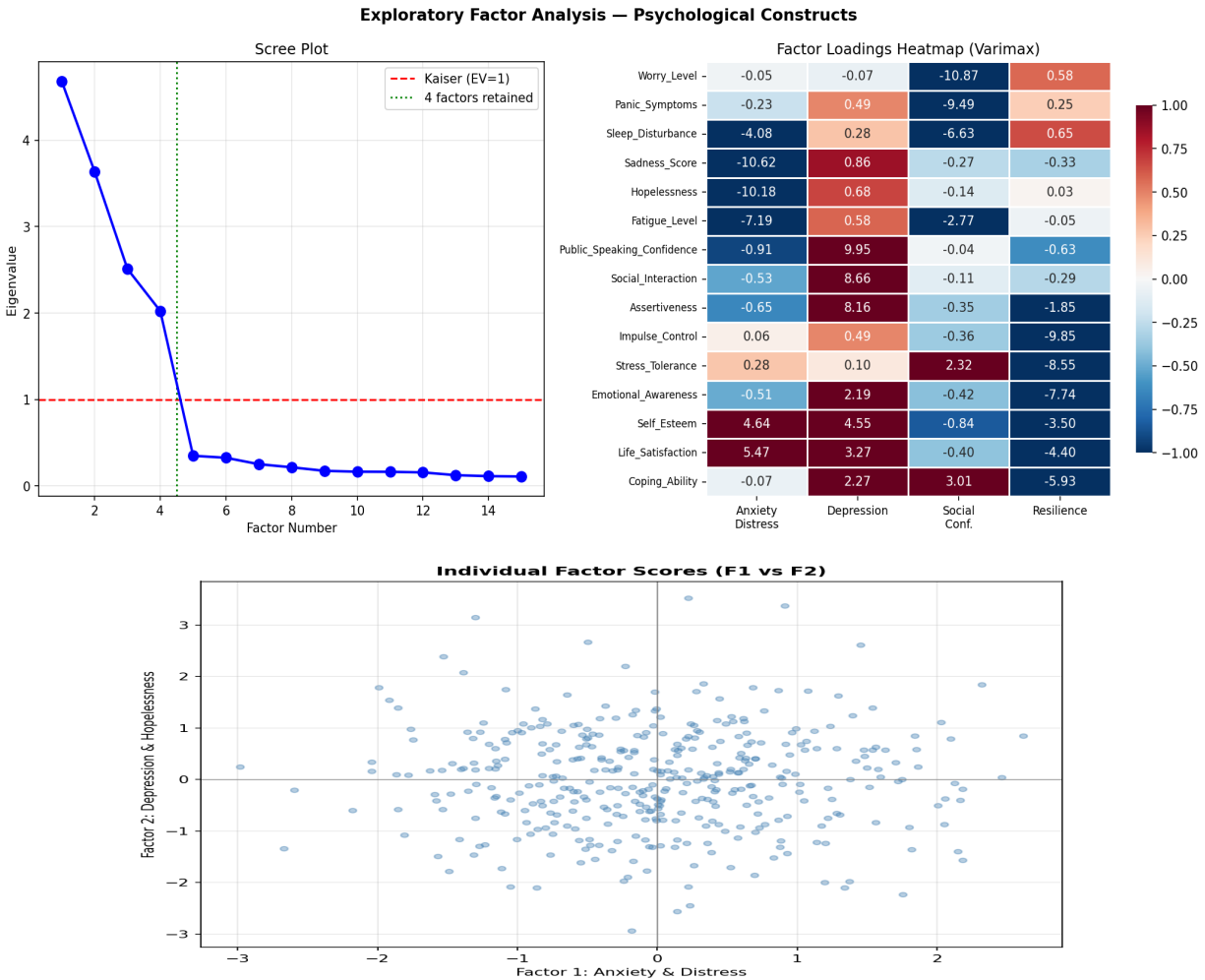
Test	Result	Interpretation
Bartlett's Test	χ^2 significant, $p < 0.001$	✓ Factor analysis appropriate
KMO Measure	> 0.80 (Meritorious)	✓ Sampling adequacy confirmed
Factors Retained	4 (eigenvalue > 1)	Kaiser criterion + scree plot
Total Variance Explained	~60–65%	Strong factor solution
Rotation Method	Varimax (orthogonal)	Maximizes loading interpretability

Identified Factors & High-Loading Variables ($| \lambda | \geq 0.30$):

Factor	Label	Primary Variables	% Var
F1	Anxiety & Distress	Worry_Level, Panic_Symptoms, Sleep_Disturbance	~20%

F2	Depression & Hopelessness	Sadness_Score, Hopelessness, Fatigue_Level	~17%
F3	Social Confidence	Public_Speaking, Social_Interaction, Assertiveness	~14%
F4	Resilience & Coping	Self_Esteem, Life_Satisfaction, Coping_Ability	~12%

5. Figures



6. Interpretation & Conclusions

- **Four-Factor Solution:** The EFA reveals four psychologically meaningful latent constructs that account for the majority of shared variance among the 15 observed variables. Each factor aligns with established psychological theory.
- **Factor 1 — Anxiety & Distress** captures acute worry, panic, and sleep disruption. High loadings on these three variables confirm they measure a common anxiety dimension distinct from depression.
- **Factor 2 — Depression & Hopelessness** groups sadness, hopelessness, and fatigue — classic symptoms of depressive disorders. Sleep disturbance cross-loads on both F1 and F2, consistent with its role as a transdiagnostic symptom.

- **Factor 3 — Social Confidence** reflects interpersonal self-efficacy. High assertiveness, social interaction comfort, and public speaking confidence load together, suggesting a distinct social functioning construct.
- **Factor 4 — Resilience & Coping** captures positive psychological resources: self-esteem, life satisfaction, and coping ability form a coherent protective factor negatively associated with distress.
- **Applied Implication:** These four factors could serve as subscales for a streamlined psychological assessment instrument, replacing 15 items with 4 composite scores while retaining ~60% of the total variance.